



Roll Number

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026

Even Semester - 2025-26
M(IT)201 - ENGINEERING MATHEMATICS - II

TIME ALLOTTED: 3Hours

Maximum Marks : 70

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. The differential equation $(ax+by+c)dx + (Ax+By+C)dy = 0$ is exact if and (1) CO1 BL1 PO1
only if
(a) $a=B$ (b) $b=A$
(c) $a=b, A=B$ (d) none
2. The general form of a Clairaut's equation is (1) CO1 BL1 PO1
(a) $y=-px+f(p)$ (b) $y= 2px+f(p)$
(c) $y=px+p$ (d) $y= px+f(p)$
3. The Bernoulli equation is given by: (1) CO1 BL1 PO1
(a) $dy/dx+P(x)y=Q(x)y^n$ (b) $y''+P(x)y'+Q(x)y=0$
(c) $y'+P(x)y=Q(x)$ (d) None of these
4. The C.F of the equation $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} = 3x$ is (1) CO2 BL3 PO1
(a) $c_1x+c_2e^{3x}$ (b) $c_1e^x+c_2e^{3x}$
(c) $c_1+c_2e^{3x}$ (d) none
5. $\{1/(D^2+1)\}(\cos 2x) = ?$ (1) CO2 BL2 PO1
(a) $(1/3)\cos 2x$ (b) $(-1/3) \cos 2x$
(c) $(-1/3) \cos 2x+c$ (d) $(-1/3) \sin 2x$

6. The Wronskian is most commonly used while finding the general solution of a differential equation by: (1) C01 BL1 PO1
- (a) Separation of variables (b) Variation of parameters
(c) Exact equation method (d) Laplace transform
7. $L^{-1}\left[\frac{5}{(s+2)^2}\right]$ is equals to (1) C02 BL2 PO1
- (a) $5te^{-2t}$. (b) $5e^{-2t}$.
(c) t^2e^{-2t} . (d) e^{-2t} .
8. The Laplace transform of $\sin(bt)$ is: (1) C01 BL1 PO1
- (a) s/s^2+b^2 (b) b/s^2+b^2
(c) $1/s^2+b^2$ (d) $b/s+b$
9. If $F(s)$ is the Laplace transform of $f(t)$, then the Laplace transform of $e^{-at}f(t)$ is: (1) C01 BL1 PO1
- (a) $F(s+a)$. (b) $F(s-a)$.
(c) $e^{as}F(s)$. (d) $e^{-as}F(s)$.
10. $(\Delta - \nabla)x^2$ is equal to (notations have their usual meaning) (1) C02 BL2 PO1
- (a) h^2 . (b) $-2h^2$.
(c) $2h^2$. (d) h .
11. The number of significant digits is 0.08340 is (1) C02 BL2 PO1
- (a) 3. (b) 4.
(c) 5. (d) 6.
12. Newton's Backward Interpolation formula is used, when the interpolating point is (1) C01 BL2 PO1
- (a) near the beginning of the table (b) near the end of the table
(c) near the middle of the table (d) all the above

Part B**(Answer any three of the following) 3x5=15**

13. Solve: $e^x \sin y \, dx + (e^x + 1) \cos y \, dy = 0$ (5) C03 BL3 PO1
14. Solve: $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = (x+1)e^x$. (5) C02 BL2 PO1
15. Find missing term in the following tables (5) C03 BL3 PO1

X: 0 1 2 3 4
F(x): 1 3 9 -- 81

16. Evaluate $L^{-1}\left\{\frac{6+s}{s^2+6s+13}\right\}$. (5) C02 BL2 PO1
17. Write down the approximate representation of $2/3$ correct upto four significant digits and find the absolute, relative and percentage error. (5) C04 BL1 PO1

Part C
(Answer any three of the following) 3x15=45

18. (a) Find the general and singular solution of $yp^2 - 2xp + y = 0$ with substitutions: $x=u$ and $y^2=v$ to reduce Clairaut's form. (8) C04 BL4 PO1
- (b) Solve: (7) C03 BL3 PO1
- $$(x+y+1)\frac{dy}{dx} = 1$$
19. (a) Use Simpson's one third rule to evaluate $\int_0^6 \frac{dx}{(1+x)^2}$, taking six equal sub-intervals of $[0,6]$, correct to three decimal places. (7) C04 BL2 PO1
- (b) Find the Laplace transform of $4\cos^2 2t$ and $e^{-t} \sin^2 t$ (8) C03 BL2 PO1
20. (a) Compute the values of $f(3.5)$ and $f(7.5)$ using Newton's interpolation from the following table: (8) C04 BL3 PO1
- | | | | | | | | | | | |
|----|-----|-----|-----|---|---|---|-------|----|----|-----|
| x: | 3 | 4 | 5 | 6 | 7 | 8 | f(x): | 27 | 64 | 125 |
| | 216 | 343 | 512 | | | | | | | |
- (b) Apply the method of variation of parameters to solve: (7) C02 BL2 PO1
- $$\frac{d^2 y}{dx^2} - 3\frac{dy}{dx} + 2y = 9e^x$$
21. (a) Find: (8) C03 BL3 PO1
- $$\int_0^\infty t e^{-2t} \sin t \, dt$$
- (b) Using convolution theorem find the ILT of the function (7) C03 BL3 PO1
- $$\frac{1}{(s-2)(s^2+1)}$$

22.

- (a) Given that $\frac{dy}{dx} = x + 2y$, with the initial condition $y = 1$ for $x = 0$. Find $y(0.1)$ correct up to four decimal places, taking the step length 0.05, using Euler's method. (7) C02 BL2 PO1
- (b) Using the fourth-order Runge-Kutta method, find $y(0.2)$ given that $\frac{dy}{dx} = 1 + (y + 1)^2$, $y(0) = 0$, taking $h = 0.2$, correct up to 4 decimal places. (8) C02 BL2 PO1